

Polymorphism for Operators

Python also exhibits polymorphism with operators. Consider the following example with the + operator:

```
value1 = 4
value2 = 5
result = value1 + value2
print(result)
```

which prints:

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The + operator here clearly means “add” in a mathematic sense because both variables contain integer values. But now consider this second example:

```
value1 = 'Joe'
value2 = 'Schmoe'
result = value1 + value2
print(result)
```

which prints:

JoeSchmoe

The line `result = value1 + value2` is exactly the same as in the first example, but it performs a completely different operation. With string values, the + operator performs a string concatenation. The same operator was used, but a different action was performed.

This technique of having multiple meanings for an operator is commonly known as *operator overloading*. For some classes, the ability to overload operators adds highly useful features and greatly improves the readability of client code.